

Pre-training for Speech Translation: CTC Meets Optimal Transport

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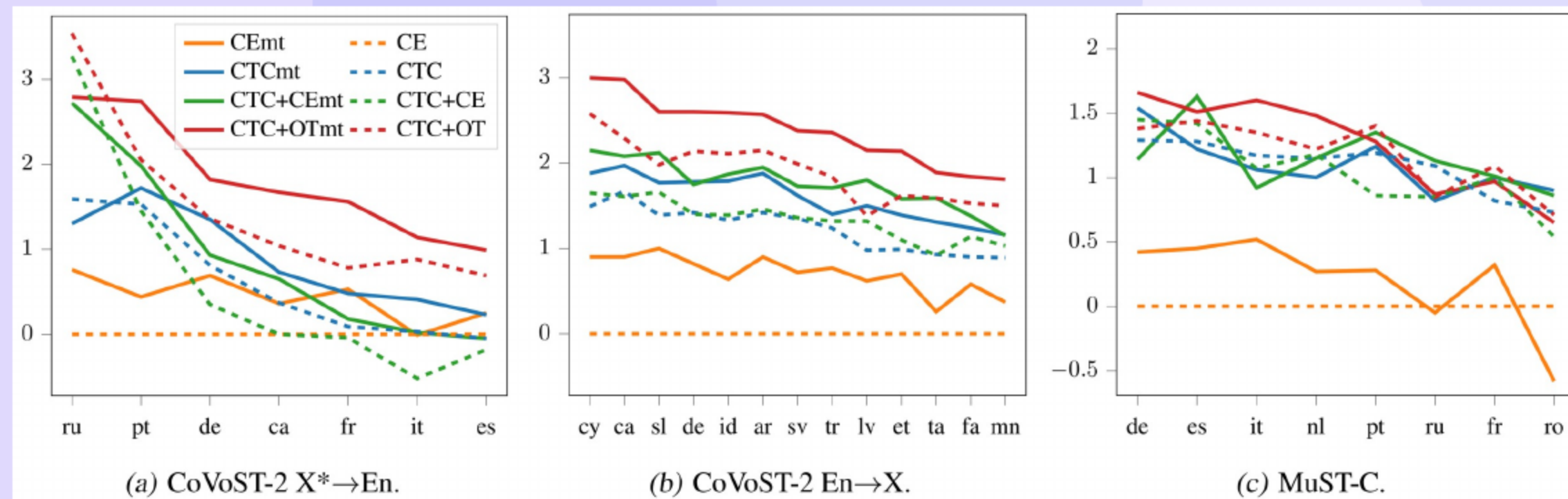
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Title and Authors

- Pre-training for Speech Translation: CTC Meets Optimal Transport
- Authors from Univ. Grenoble Alpes and Meta AI
- Phuong-Hang Le, Hongyu Gong, Changhan Wang, Juan Pino, Benjamin Lecouteux, Didier Schwab

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Why CTC and OT?



- Standard speech translation pre-training faces a modality gap between speech and text
- CTC pre-training outperforms cross-entropy for downstream speech translation by encoding alignments in the encoder
- Optimal transport aligns acoustic and text representations and enables MT decoder reuse
- CTC with optimal transport reaches state-of-the-art without external data

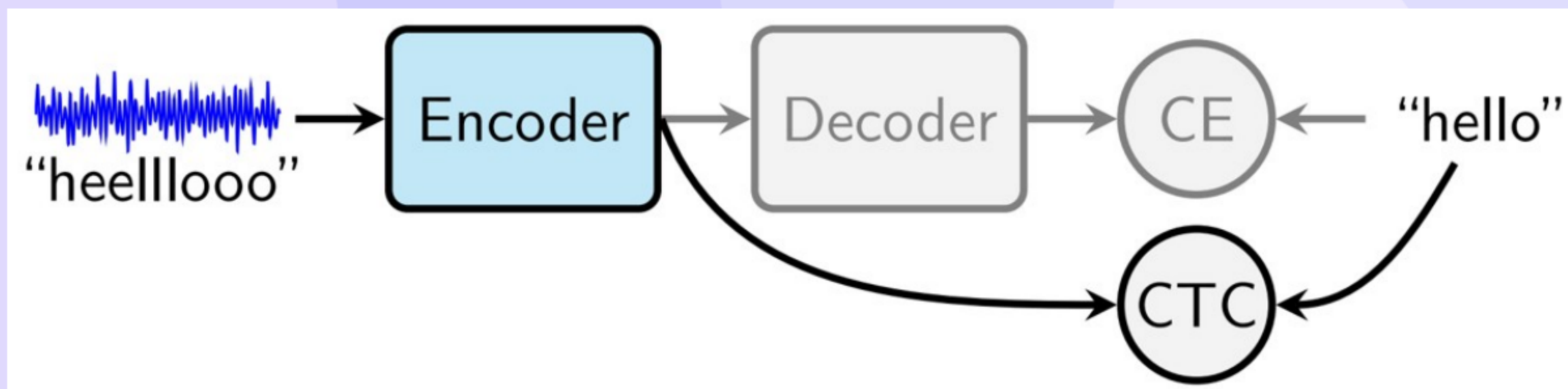
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Positioning and Comparisons

- Siamese pre-training is generic and requires no architecture changes to speech translation
- Prior gap-bridging modifies models or forces length matching and adds complexity
- Optimal transport naturally handles unequal lengths and provides principled alignment
- Replacing cross-entropy with CTC benefits recent multi-task systems
- CTC vs CE shows quantitative gains; OT strongest in comparisons

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CTC Learns Alignments



- Connectionist temporal classification learns frame to token alignments without a decoder
- Blank handling collapses repeated labels and stabilizes **alignment learning**
- Summation over alignments embeds timing cues directly in the encoder states
- Encoder-side alignments reduce the speech to text **transfer gap**

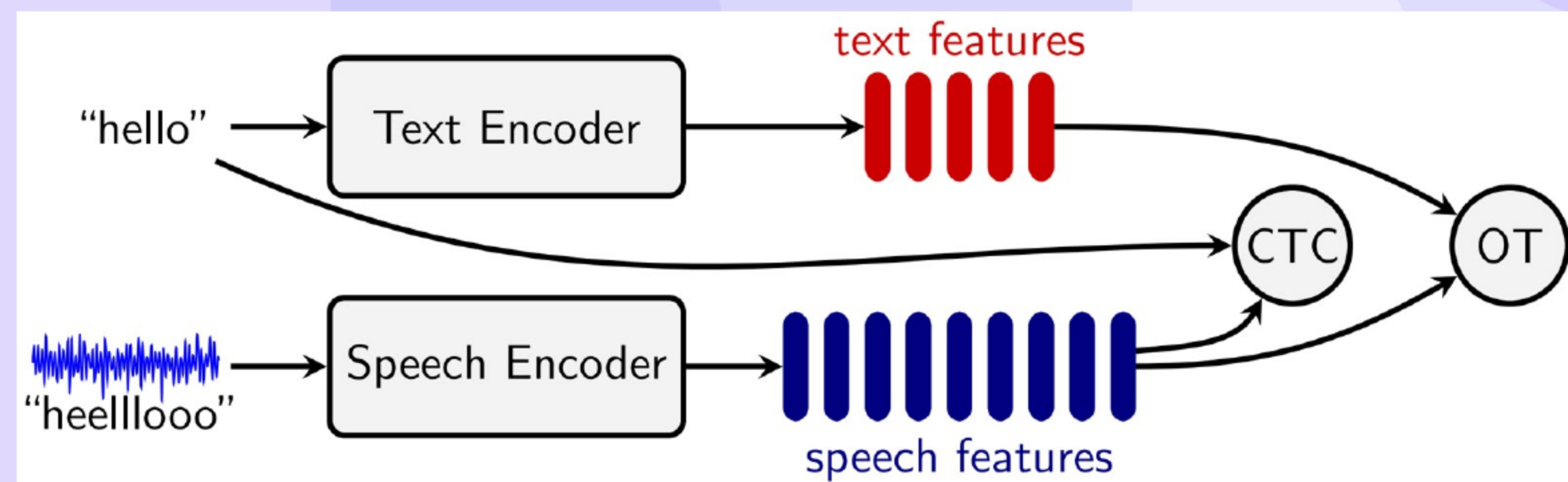
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CTC vs CE: Why

- CTC improves speech translation even when cross-entropy yields **lower WER** for ASR
- Cross-entropy advantages reside in the decoder that is later discarded for speech translation
- CTC alone cannot recover alignments from the missing machine translation encoder
- Adding optimal transport restores cross-modal structure and **boosts BLEU**
- On MuST-C dev, CTC+OT > CTC > CE for ST BLEU (Table 1)

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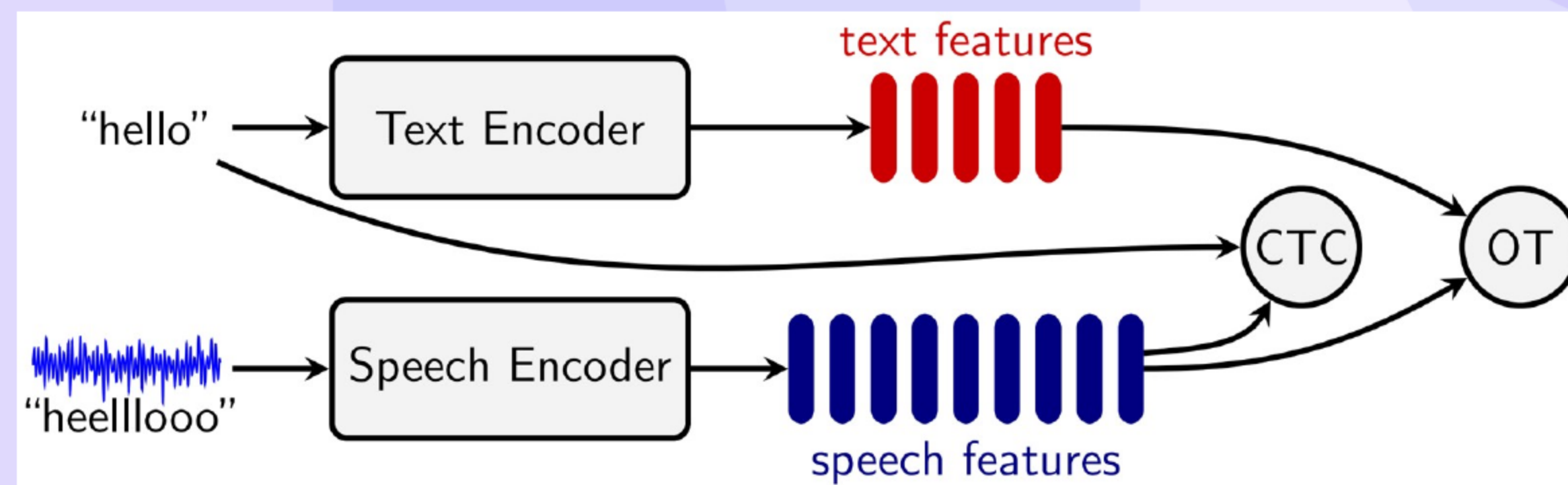
Method Overview



- Siamese pre-training aligns a CTC speech encoder with a text encoder using optimal transport
- Optional initialization of the text encoder from machine translation improves **representation quality**
- Speech translation fine-tuning reuses the MT decoder and discards the auxiliary text encoder
- Inference uses only the speech encoder and the machine translation decoder

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OT Intuition



- Optimal transport minimizes a transport cost between speech and text encoder sequences
- Entropic regularization with Sinkhorn provides **efficient** and differentiable training
- Sequence length mismatch is handled natively without explicit alignment heuristics
- Joint training with CTC preserves **ASR alignment** while improving cross-modal closeness

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Positional OT Recipe

- Appending normalized positions with a tunable weight biases toward monotonic alignments
- The framework works with several alternatives but optimal transport performs **consistently best**
- Final recipe favors no weight sharing and uses positional features during alignment
- Initialization from a machine translation encoder further **improves transfer**
- No sharing + positional yields best dev BLEU (Table 3)

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Datasets

- MuST-C covers English to eight European targets with topic-diverse TED talks
- CoVoST-2 spans many directions with crowd-sourced accents and **imbalanced** resources
- Standardized splits enable controlled ablations and multilingual evaluation

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Experimental Setup

- Multilingual focus with bilingual ablations on German and French for cost control
- FAIRSEQ S2T implementation with 12 encoder and dual 6-layer decoders
- Model width uses medium for analysis and large for **best results**
- Development splits guide selection before final test evaluation
- Large model (d=1024) used for final SOTA runs

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Ablations and Trends

- CTC improves speech translation BLEU over cross-entropy despite **higher WER**
- Optimal transport adds further gains with similar training cost to combined losses
- Optimal transport outperforms Euclidean, adversarial, and KL under sequence length mismatch
- No weight sharing and positional encoding emerge as the **best** configuration
- Multilingual dev avg: CE 25.42 \u2192 CTC+OT 26.67 (+1.25)

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Comparative Results

- On MuST-C test, Siamese pre-training with CTC and optimal transport achieves the **highest** average BLEU
- On CoVoST-2, consistent gains appear in English to many targets and many sources to English
- The method complements multi-task training and improves strong baselines
- Results surpass prior no-external-data systems by a **large margin**
- MuST-C avg: CE 28.8 \u2192 CTC+OT 29.8; +3.2 over Adapters 26.6

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Key Takeaway

- Bridges the speech–text modality gap
- Enables MT-decoder reuse; no model changes
- State-of-the-art without external data

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Limitations & Impact

- Low-resource directions remain challenging.
- Privacy/fairness require safeguards.

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Open Questions

- Scaling with external data and larger models may yield **further gains**
- Integrating optimal transport as an auxiliary loss in multi-task training could enhance alignment
- Robustness under domain shift, noisy transcripts, and disfluencies remains **uncertain**
- Learned positional costs and soft dynamic time warping hybrids may improve **efficiency**

Thank You

Questions & Discussion